

External Shock Transmission to Hong Kong Under the Currency Board: Macro-Econometric Evidence from VAR and VECM

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May 11, 2026

Abstract

This note asks how US monetary conditions and China growth shocks propagate to Hong Kong's real economy under the currency board. I estimate a quarterly macro-econometric system for 1998Q1–2026Q1 using official Hong Kong data, US federal funds rates, China real GDP growth, and Hong Kong exports to the Chinese Mainland. The baseline is a seven-variable external-first VAR selected by BIC; a VECM is used as a long-run robustness check because the data show cointegration among the nonstationary variables.

The current baseline results give a simple interpretation. China real activity and the Hong Kong export channel matter for real activity: at the eight-quarter horizon, exports to the Chinese Mainland explain 24.9% of Hong Kong GDP growth forecast error variance, while China GDP explains 16.1%. The US channel is clearest in interest rates: the federal funds rate explains 53.7% of HIBOR variance and 10.6% of unemployment variance. A rank-2 VECM strengthens the long-run US rate channel, assigning 65.2% of HIBOR variance and 47.0% of unemployment variance to the US FFR shock. These results are best read as reduced-form macro-econometric evidence, not as clean causal identification. Residual autocorrelation remains in several VAR equations, so the strongest use of the project is as a research note and disciplined forecasting/diagnostic framework.

Keywords: Hong Kong, currency board, VAR, VECM, FEVD, monetary transmission, China spillovers, exports

JEL Classification: E32, E52, F41, F42

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1 Introduction

Hong Kong’s macroeconomic environment is shaped by two external anchors. The first is monetary: under the Linked Exchange Rate System, Hong Kong imports US monetary conditions through the HKD–USD currency board. The second is real: Hong Kong is tightly linked to mainland China through trade, re-exports, services, and finance. The empirical question is therefore direct: when the US rate cycle and China growth cycle move, which channel shows up in Hong Kong’s real economy?

This note studies that question with a quarterly VAR/VECM framework. The baseline system contains seven variables ordered external-first:

```
us_ffr, china_gdp, hk_exports_china_yoy,  
gdp_growth, cpi_inflation, unemployment, hibor_3m.
```

This ordering treats the US rate and China activity variables as externally given within the quarter, allows China activity to affect Hong Kong exports, and then traces the propagation to Hong Kong GDP, inflation, unemployment, and HIBOR.

The main findings are:

1. **China propagates through trade and real activity.** In the current VAR(1), exports to the Chinese Mainland explain 24.9% of Hong Kong GDP growth variance at the eight-quarter horizon. China real GDP growth explains another 16.1%. This is now a cleaner macro story than the earlier version because the model includes an explicit China-to-Hong-Kong trade channel.
2. **The US channel is mainly an interest-rate channel.** The US FFR explains 53.7% of HIBOR variance in the baseline VAR. It also explains 10.6% of unemployment variance, but only 1.2% of GDP growth variance at the same horizon.
3. **VECM strengthens long-run currency-board transmission.** Johansen diagnostics select rank 2 for the I(1) subset. In the rank-2 VECM, the US FFR explains 65.2% of HIBOR variance and 47.0% of unemployment variance. This supports the idea that differenced VARs can understate the long-run interest-rate comovement created by the peg.
4. **The model is useful, but not finished as causal evidence.** Residual autocorrelation remains in several VAR equations. The results should therefore be described as macro-econometric transmission evidence, not as a fully identified structural causal design.

The contribution is practical rather than methodological. The note builds a transparent quarterly system for Hong Kong, documents the data sources, and separates the US

rate channel from the China real-activity/export channel. The results are useful for understanding why fiscal and macroprudential instruments matter in Hong Kong: monetary conditions are imported, while major real shocks come from external demand that cannot be offset by local interest-rate policy.

2 Literature Review

This note connects three literatures. First, the small-open-economy VAR literature studies how foreign interest rates and foreign output transmit to domestic macro variables (Cushman and Zha, 1997; Kim, 2001). The external-first recursive ordering used here follows that logic. Second, work on sign-restricted VARs emphasizes that recursive timing assumptions are only one way to discipline structural interpretation (Uhlig, 2005; Rubio-Ramírez et al., 2010). This project includes sign-restriction diagnostics, but the baseline claims remain reduced-form. Third, currency-board macroeconomics emphasizes the loss of domestic monetary autonomy under a hard peg (Hanke, 2002). For Hong Kong, this means the FFR–HIBOR relationship is not an incidental correlation; it is part of the monetary regime.

The Hong Kong setting adds a China channel. Prior work on Hong Kong monetary transmission stresses the importance of US rates for local financial conditions (Genberg and He, 2009), while work on mainland-Hong Kong integration emphasizes the increasing role of China in Hong Kong’s business cycle (He et al., 2014). This note puts both channels in the same empirical system and adds exports to the Chinese Mainland as an explicit propagation variable.

3 Data

3.1 Variables and Sources

The current canonical dataset covers 1998Q1–2026Q1, with 113 raw quarterly observations. The transformed estimation sample has 112 observations because some variables enter in first differences.

Table 1: Variable Definitions and Sources

Variable	Description	Unit	Source
<code>us_ffr</code>	US effective federal funds rate	%	FRED FEDFUNDS
<code>china_gdp</code>	China real GDP growth	YoY %	OECD QNA B1GQ, GY
<code>hk_exports_china_yoy</code>	HK exports to Chinese Mainland	YoY %	C&SD 410-50013
<code>gdp_growth</code>	HK real GDP growth	YoY %	C&SD 310-30001
<code>cpi_inflation</code>	HK Composite CPI inflation	YoY %	C&SD 510-60001
<code>unemployment</code>	HK unemployment rate	%	C&SD 210-06101
<code>hibor_3m</code>	3-month HIBOR	%	HKMA API

The Hong Kong series come from official Census and Statistics Department (C&SD) or HKMA sources. CPI uses the official Composite CPI year-on-year monthly series from C&SD table 510-60001, aggregated to quarterly means. The China variable is documented in the current metadata as OECD QNA real GDP YoY growth. The code keeps the column name `china_gdp` for compatibility; if the OECD fetch fails in a future rebuild, the fetcher can fall back to FRED nominal GDP, in which case interpretation must be changed to “China nominal activity.” The current snapshot uses the real-activity interpretation recorded in `data/source_metadata.json`.

3.2 Stationarity and Cointegration

I use ADF and KPSS tests jointly. Variables classified as unit-root by the diagnostic rule are first-differenced for the VAR/BVAR. In the current run, `china_gdp`, `cpi_inflation`, and `unemployment` enter as first differences. The remaining variables enter in levels: `us_ffr`, `hk_exports_china_yoy`, `gdp_growth`, and `hibor_3m`. The HIBOR tests conflict, so the conservative project rule keeps it in levels.

Johansen diagnostics on the $I(1)$ subset select rank 2 at the 95% trace-test rule after the project cap. The VECM robustness specification therefore uses rank 2 with a restricted constant. This is economically plausible in a currency board setting, but the exact VECM interpretation remains sensitive to rank and deterministic choices.

4 Econometric Framework

4.1 Estimand and Design

The estimand is dynamic propagation: the forecast-error contribution of US monetary conditions, China real activity, and the Hong Kong-China export channel to Hong Kong GDP growth, CPI inflation, unemployment, and HIBOR. The baseline design is a reduced-form quarterly VAR with recursive identification for impulse responses and FEVD. Identification relies on a small-open-economy ordering assumption: US and China variables do not respond contemporaneously to Hong Kong macro shocks.

4.2 VAR Specification

The reduced-form VAR(p) is

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \quad u_t \sim (0, \Sigma_u),$$

where y_t is the seven-variable vector listed above. BIC selects $p = 1$ with a low parameter-observation ratio of 0.071. The fitted VAR is stable, with maximum companion eigenvalue modulus 0.9389. GFC and COVID pulse dummies are included in the main fit to absorb large event shocks, but residual autocorrelation remains in several equations and is treated as a limitation.

4.3 Cholesky Identification

For FEVD and impulse-response analysis, I decompose $\Sigma_u = PP'$ using the external-first ordering. The ordering is

$$\begin{aligned} &\text{us_ffr, china_gdp, hk_exports_china_yoy,} \\ &\text{gdp_growth, cpi_inflation, unemployment, hibor_3m.} \end{aligned}$$

This is a timing restriction, not a full policy-invariant structural model.

4.4 VECM Robustness

The VECM is

$$\Delta y_t = \alpha \beta' y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + c + u_t.$$

It is estimated on level data to preserve long-run relations. In the current configuration, the VECM uses rank 2 and one lag of differenced terms. It is reported as long-run robustness because the currency board implies persistent FFR–HIBOR comovement.

4.5 FEVD

The forecast error variance decomposition is

$$\text{FEVD}_{i,j}(h) = \frac{\sum_{s=0}^h \theta_{ij,s}^2}{\sum_{s=0}^h \sum_{k=1}^K \theta_{ik,s}^2}.$$

The paper reports the eight-quarter horizon because it is long enough to show propagation while still close to the business-cycle question.

5 Results

5.1 Baseline Diagnostics

The canonical run uses `python hk_var_model.py -lag-criterion bic -model-type var`. The sample is 1998Q1–2026Q1, BIC selects one lag, and the fitted VAR is stable. The main diagnostic weakness is residual autocorrelation: Ljung–Box tests at lag 8 still reject in the US FFR, China GDP, exports-to-China, GDP-growth, and CPI equations. This does not invalidate the project as a research note, but it limits strong causal or publication-style claims.

5.2 Granger-Causality Evidence

Reduced-form Granger tests line up with the intended channels. China GDP Granger-causes exports to the Chinese Mainland, Hong Kong GDP growth, CPI inflation, and unemployment in the transformed system. The US FFR Granger-causes Hong Kong GDP growth, unemployment, and HIBOR. These tests are predictive checks, not structural identification, but they support keeping both external channels in the same system.

5.3 Forecast Error Variance Decomposition

Table 2 reports the central FEVD results at the eight-quarter horizon. The VAR columns show the baseline seven-variable model. The VECM columns show the long-run robustness comparison for the US and China shocks.

Table 2: Forecast Error Variance Decomposition at $h = 8$ Quarters

Target	VAR(1)			VECM rank 2	
	US FFR	China GDP	HK exports	US FFR	China GDP
HK GDP growth	1.2%	16.1%	24.9%	4.2%	21.9%
HK CPI inflation	0.5%	6.6%	4.0%	20.1%	43.2%
HK unemployment	10.6%	6.3%	5.4%	47.0%	8.6%
3M HIBOR	53.7%	0.5%	10.1%	65.2%	0.5%

Three results matter. First, the China channel is now best understood as a combination of China real activity and exports to the Chinese Mainland. For HK GDP growth, the direct China GDP shock explains 16.1%, while the export-channel shock explains 24.9%. Second, the US channel is concentrated in HIBOR: the FFR shock explains 53.7% of HIBOR variance in the VAR. Third, the VECM strengthens the long-run US channel, especially for unemployment and HIBOR. This is consistent with the currency-board mechanism, but it should be treated as robustness rather than a standalone causal estimate.

The exports equation itself is also informative. At the same horizon, own export shocks explain 77.0% of exports-to-China variance, while China GDP explains 15.5%. This suggests that the exports series contains both China-demand content and Hong Kong-specific trade volatility.

5.4 Forecast Comparison

In level-space expanding-window forecast comparison, the VECM wins 18 of 28 RMSE cells against the VAR, while the VAR wins 10. The VECM performs better for US FFR, China GDP, CPI, and longer-horizon unemployment forecasts. The VAR is better for exports to China and some GDP/HIBOR horizons. This mixed pattern is exactly why the paper keeps the VAR as the transparent baseline and the VECM as long-run robustness.

5.5 Sign Restrictions and TVP-VAR

The sign-restriction block accepts 500 rotations out of 50,000 candidate draws under the current restrictions. Accepted responses are qualitatively consistent with the economic story: US tightening raises HIBOR and weakens real activity, while positive China activity shocks support Hong Kong output and prices. These results are useful as a plausibility check but do not replace the recursive baseline.

The TVP-VAR block is exploratory. Current residual-standard-deviation checks do not show it dominating the constant-parameter VAR, so it should not be a main claim.

Its role is to flag possible structural change around the GFC, COVID, and the recent Fed tightening cycle.

6 Robustness and Limitations

6.1 Ordering and Subsamples

Ordering robustness and sub-sample checks preserve the broad interpretation: China and external-demand variables matter more for real activity, while US rates matter more for HIBOR. The post-GFC period has different coefficient norms, which is unsurprising given repeated large shocks and shorter effective sample length.

6.2 Main Limitations

The main limitations are straightforward:

1. The model is reduced-form. Recursive and sign restrictions impose identifying assumptions, but this is not an external-instrument monetary shock design.
2. Residual autocorrelation remains under the BIC-selected VAR(1). More lags or BVAR shrinkage can be used as robustness, but they increase model complexity.
3. The China variable must be checked through metadata after each data rebuild. In the current snapshot it is OECD real GDP growth; if the fallback triggers, it becomes nominal activity and the paper language must change.
4. Property prices and high-frequency US monetary shock data are useful sidecar extensions, but they are not part of the current baseline.

7 Conclusion

This note gives a disciplined answer to the research question. Under the Hong Kong currency board, US monetary conditions propagate most clearly through the interest-rate channel: the US FFR explains over half of HIBOR variance in the baseline VAR, and the VECM raises this share further. China real activity and exports to the Chinese Mainland matter more for the real economy: together they account for a substantial share of Hong Kong GDP-growth variance at the eight-quarter horizon.

The clean interpretation is therefore not “US versus China” in a single ranking. It is channel-specific. US shocks dominate the monetary/interest-rate channel. China growth and the export channel dominate the real-activity story. Because Hong Kong imports monetary conditions but faces real shocks from China, stabilization necessarily relies on fiscal, prudential, and structural policy rather than independent monetary policy.

The project is now suitable as a macro-econometric research note. The next best extensions are narrow: use a shorter-sample monetary-surprise series for US shock robustness, or add property prices only if the question becomes explicitly macro-financial.

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